AI-ASSISTED CODING

ASSIGNMENT-19.1

Code Translation: Converting between programming languages

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BATCH NO: 05

Task 1.1 :

You are part of a multinational development team. The backend is written in
Java, but a new module requires a Python implementation of the same algorithm for integration with a data science pipeline.
• Task 1: Use AI-assisted coding to translate a given Java bubble sort program into Python. Verify that the translated code works correctly.

Prompt:

Translate the given Java bubble sort program into Python and verify that it works correctly.

CODE GENERATED:

Python:

```python
def bubble_sort(arr):
    n = len(arr)
    for i in range(n - 1):
        for j in range(n - i - 1):
            if arr[j] > arr[j + 1]:
                # swap
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr
```

```
# Test the function
numbers = [5, 3, 8, 4, 2]
sorted_numbers = bubble_sort(numbers)
print("Sorted array:", sorted_numbers)
```

OUTPUT:

```
Sorted array: [2, 3, 4, 5, 8]
```

OBSERVATION :

The translated Python program successfully sorts the list of numbers in ascending order. The output matches the result of the original Java bubble sort program.

Task 1.2:

introduce errors in the Python version to check if the input list is empty or contains non-numeric values

PROMPT:

Introduce errors in the Python bubble sort program to check if the input list is empty or contains non-numeric values.

CODE:

```
def bubble_sort(arr):
    # Check if list is empty
    if len(arr) == 0:
```

```python
        print("Error: The list is empty!")
        return None

    # Bubble sort logic (contains intentional mistake for testing)
    for i in range(len(arr)):              # Intentional mistake
        for j in range(len(arr) - i):      # Intentional mistake
            if arr[j] > arr[j + 1]:        # Causes TypeError for non-numeric
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr


# Test cases
print(bubble_sort([]))              # Empty list → should show error
print(bubble_sort([3, "a", 1]))     # Non-numeric → TypeError
```

OUTPUT:

```
Error: The list is empty!
None
```

OBSERVATION :

The introduced errors successfully demonstrate that the program can detect and respond to invalid inputs like empty lists and non-numeric values.

TASK -2.1:

A company's legacy codebases stores and processes files in C++, but the analytics team needs an equivalent program in JavaScript (Node.js) for

integration with a web dashboard.
• Task: Translate a given C++ file read-and-write program into JavaScript using AI assistance. Ensure the script reads a text file and writes processed output to a new file.

PROMPT:

Translate the given C++ program that reads data from a file and writes processed content to another file into an equivalent JavaScript (Node.js) program.

CODE GENERATED:

```
// Import the file system module
const fs = require('fs');

// Read content from input.txt
const data = fs.readFileSync('input.txt', 'utf8');

// Process content
const lines = data.split('\n').map(line => `Processed: ${line}`);

// Write to output.txt
fs.writeFileSync('output.txt', lines.join('\n'));

console.log('File processing completed.');
```

OUTPUT:

```
File processing completed.
```

```
input.txt
Hello World
AI and Data Science
File Handling Example
Bubble Sort Translation
End of File

```

output.txt

```
Processed: Hello World
Processed: AI and Data Science
Processed: File Handling Example
Processed: Bubble Sort Translation
Processed: End of File
Processed:
```

OBSERVATION:

The JavaScript (Node.js) program successfully reads the data from input.txt, processes each line by adding “Processed: ” before it, and writes the result to output.txt.

TASK – 2.2:

Add error handling in the JavaScript version to gracefully handle missing files or permission errors.

PROMPT:

Modify the JavaScript (Node.js) file-handling program to include error handling for missing files and permission errors.

CODE GENERATED:

```javascript
const fs = require('fs');

try {
    // Try to read the file
    const data = fs.readFileSync('input.txt', 'utf8');
    const lines = data.split('\n').map(line => `Processed: ${line}`);

    // Try to write to output file
    fs.writeFileSync('output.txt', lines.join('\n'));
```

```
    console.log('File processing completed successfully.');

} catch (error) {
    // Handle possible errors
    if (error.code === 'ENOENT') {
        console.error('Error: Input file not found.');
    } else if (error.code === 'EACCES') {
        console.error('Error: Permission denied while accessing files.');
    } else {
        console.error('An unexpected error occurred:', error.message);
    }
}
```

OUTPUT:

```
Error: Input file not found.
```

OBSERVATION:

The program shows an error message if the input file is missing or if there is no permission to access the file. When the file is present, it reads and writes the data correctly without any errors.

TASK – 3.1:

News Headlines API
A news aggregator wants to display the latest technology news headlines using a news API. Sometimes, the API responds slowly or returns incomplete data.

Use AI-assisted coding to fetch the top 5 technology headlines and print them neatly in the console. Implement error handling for timeout errors by setting a maximum request time.

PROMPT:

Write a Python script (using AI assistance) to fetch the top 5 technology news headlines and print them neatly in the console. Handle timeout errors by setting a maximum request time.

CODE GENERATED:

```
import requests

def fetch_tech_news():
    url = "https://saurav.tech/NewsAPI/top-headlines/category/technology/in.json"

    try:
        # Set timeout to 5 seconds
        response = requests.get(url, timeout=5)
        response.raise_for_status()  # Raises an exception for HTTP errors

        data = response.json()

        # Check if data contains valid articles
        if "articles" in data and len(data["articles"]) > 0:
            print("\n📰 Top 5 Technology News Headlines:\n")
            for i, article in enumerate(data["articles"][:5], start=1):
                print(f"{i}. {article.get('title', 'No title available')}")
        else:
            print("⚠ No articles found or incomplete data received.")

    except requests.exceptions.Timeout:
        print("❌ Request timed out! The server took too long to respond.")
    except requests.exceptions.RequestException as e:
        print(f"❌ An error occurred: {e}")

# --- Main Program ---
fetch_tech_news()
```

OUTPUT:

```
📰 Top 5 Technology News Headlines:

1. WhatsApp Business could charge users for linking more than four devices - XDA Developers
2. Google Messages causing severe battery drain and overheating? Here’s a temporary fix - XDA Developers
3. Redmi 10 Power with 6,000mAh Battery Launched in India: Check Price and Specs - The Quint
4. Alienware Doubles Down on AMD, Launches New Ryzen Laptops and Desktop - Gizmodo
5. Apple MagSafe Battery Pack Will Now Charge Your iPhone Faster: Here’s How - News18
```

OBSERVATION:

The AI-generated script successfully fetches and displays the latest **technology headlines** from a public news API.

TASK 3.2:

Clean and preprocess the headlines by removing special characters and converting text to title case. Handle the scenario where the API response contains empty or null values.

PROMPT:

Extend the script to clean and preprocess the fetched headlines by removing special characters and converting them to title case. Handle cases where the API response contains empty or null values.

CODE GENERATED:

```python
import requests
import re

def clean_headline(text):
    """Remove special characters and convert to title case."""
    if not text or text.strip() == "":
        return "No Title Available"
    # Remove special characters except alphabets, numbers, and spaces
    cleaned = re.sub(r'[^A-Za-z0-9\s]', '', text)
    return cleaned.title().strip()

def fetch_and_clean_news():
    url = "https://saurav.tech/NewsAPI/top-
headlines/category/technology/in.json"

    try:
        response = requests.get(url, timeout=5)
        response.raise_for_status()
        data = response.json()
```

```
        if "articles" in data and len(data["articles"]) > 0:
            print("\n📰 Cleaned Top 5 Technology Headlines:\n")

            for i, article in enumerate(data["articles"][:5], start=1):
                raw_title = article.get("title", "")
                clean_title = clean_headline(raw_title)
                print(f"{i}. {clean_title}")

        else:
            print("⚠️ No valid articles found or incomplete data received.")

    except requests.exceptions.Timeout:
        print("❌ Request timed out! The server took too long to respond.")
    except requests.exceptions.RequestException as e:
        print(f"❌ An error occurred: {e}")

# --- Main Program ---
fetch_and_clean_news()
```

OUTPUT:

```
📰 Cleaned Top 5 Technology Headlines:

1. Whatsapp Business Could Charge Users For Linking More Than Four Devices  Xda Developers
2. Google Messages Causing Severe Battery Drain And Overheating Heres A Temporary Fix  Xda Developers
3. Redmi 10 Power With 6000Mah Battery Launched In India Check Price And Specs  The Quint
4. Alienware Doubles Down On Amd Launches New Ryzen Laptops And Desktop  Gizmodo
5. Apple Magsafe Battery Pack Will Now Charge Your Iphone Faster Heres How  News18
```

OBSERVATION:

The script fetches the latest technology headlines, removes unwanted special characters, and formats each title in Title Case.